

**Photon-number parity in quantum spin ice:
exact selection rules for neutron, Raman and ultrasound
spectroscopy**

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Abstract

The nearest-neighbour pseudospin Hamiltonian of quantum spin ice (QSI) is invariant under a global π rotation $\mathcal{P}$ of the pseudospins about the $\tilde{x}$ axis. We show that on the ice manifold this operation is the *charge conjugation* of the emergent compact electrodynamics: it reverses both the emergent electric and magnetic fields, so an n -photon state carries parity $(-1)^n$ and photon-number parity is exactly conserved at zero magnetic field. Three consequences follow, none of which is perturbative or model-detail dependent. (i) Spectroscopic probes split into two closed families: the emergent electric field $E = S^z$ and the linear magnetic field $\sin B$ reach only odd-photon states, whereas every bond bilinear $-\cos B$, the Fleury–Loudon Raman operator, magnetostriction and the ultrasound strain vertex $—$ reaches only even-photon states. The one-photon line and the two-photon continuum therefore appear in *disjoint* experiments. (ii) At $\mathbf{q} = 0$ every single-site magnetic operator is a constant on each electric-flux sector, so uniform magnetic probes are blind to the gauge sector to all orders and only bond operators survive; this is why ultrasound, not ESR or uniform susceptibility, is the natural $\mathbf{q} \rightarrow 0$ gauge probe. (iii) A magnetic field is $\mathcal{P}$ -odd and switches the forbidden channels on at order h^2 , providing a parameter-free field protocol that distinguishes a gauge excitation from any conventional mode. We verify the selection rule on the transport-masked FCC-32 spectrum, where every computed level is bright in exactly one channel with a complement between 10^{-25} and 10^{-35} , and we show that the rule survives the pair-flip ($J_{\pm\pm}$) term relevant to the Ce pyrochlores. In octupolar QSI the neutron dipole operator has no projection on the gauge sector at all, so the even channel $—$ Raman and ultrasound $—$ is the *only* spectroscopic access to the emergent photon, and it sees the two-photon continuum whose onset lies at twice the photon gap.

INTRODUCTION

The experimental case for quantum spin ice rests on identifying, in a measured spectrum, an excitation that is the emergent photon of a compact $U(1)$ gauge theory [1–4]. In practice this identification is made by exclusion: a low-lying feature that is not a crystal-field level, not a spinon, and not a phonon is assigned to the gauge sector. Exclusion is weak evidence. What is missing is a positive, falsifiable statement of the form “*this* probe must see the mode and *that* probe must not.”

We provide such a statement. It rests on a symmetry that is already present in every nearest-neighbour pseudospin model used to describe the Ce and Yb pyrochlores, and that has, to our knowledge, not been exploited spectroscopically: the global π rotation of the pseudospins about the $\tilde{x}$ axis,

$$\mathcal{P} : \quad S_i^z \mapsto -S_i^z, \quad S_i^\pm \mapsto S_i^\mp. \quad (1)$$

$\mathcal{P}$ is one of the three $\mathbb{Z}_2$ elements of the Klein group of any XYZ model with real couplings, and is therefore exact for the full dipole–octupole Hamiltonian, not only for the XXZ limit. Its gauge-theoretic content is what matters here. In the standard dictionary [1] the emergent electric field is $E_i = S_i^z$ and the emergent vector potential is defined through $S_i^+ = e^{iA_i}$, so Eq. (1) reads

$$\mathcal{P} : \quad E \mapsto -E, \quad A \mapsto -A, \quad B \mapsto -B. \quad (2)$$

This is charge conjugation. Because the photon field is C -odd, an n -photon state has $\mathcal{P}$ eigenvalue $(-1)^n$: **photon-number parity is an exactly conserved quantum number of quantum spin ice at zero field**. The statement is the lattice analogue of Furry’s theorem in QED [5], and it is exact — not a leading-order result of the ring-exchange expansion, and not tied to any particular flux background.

The spectroscopic payoff is immediate and is the subject of this paper. The operators that experiments actually measure fall into two parity classes, and the classes cannot talk to each other. We state the classification in Sec. , verify it numerically in Sec. on finite-cluster spectra from which the torus artifacts have been removed [6], and translate it into a menu of experiments in Sec. . Section gives the field protocol that turns the selection rule into a positive test rather than a null one.

THE SELECTION RULE

$\mathcal{P}$ on the ice manifold

Write the nearest-neighbour pseudospin Hamiltonian in the exchange-diagonal local frame,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + \text{h.c.}) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + \text{h.c.}), \quad (3)$$

with $J_{zz} > 0$. Under Eq. (1) the Ising term is invariant, the $J_{\pm}$ term maps to its Hermitian conjugate, and the $J_{\pm\pm}$ pair-flip term maps to its Hermitian conjugate as well. Hence $[\mathcal{P}, H] = 0$ for the full XYZ model. In terms of the principal-axis components, $\mathcal{P}$ acts as $S^{\tilde{x}} \mapsto S^{\tilde{x}}$, $S^{\tilde{y}} \mapsto -S^{\tilde{y}}$, $S^{\tilde{z}} \mapsto -S^{\tilde{z}}$, i.e. it is the π rotation about $\tilde{x}$. The other two Klein elements are not independent on the ice manifold: $R_{\tilde{z}}(\pi) = \exp[i\pi \sum_i (S_i^z + \frac{1}{2})]$ is a c -number on any fixed- S_{tot}^z sector, so $R_{\tilde{y}}(\pi) \equiv \mathcal{P}$ there up to a phase. *One* nontrivial $\mathbb{Z}_2$ acts on the gauge sector, and it is $\mathcal{P}$.

Two structural remarks are needed before the rule can be used.

First, $\mathcal{P}$ reverses the total polarization $\mathbf{p} = \sum_i S_i^z \mathbf{r}_i$ and therefore maps the electric-flux (transport) sector $\mathcal{T}$ to $-\mathcal{T}$. It is a symmetry *within* a block only for self-conjugate sectors, $\mathcal{T} = -\mathcal{T}$. On FCC-32 the ground state of the masked theory lies in the unique $\mathcal{T} = 0$ sector, so $\mathcal{P}$ is a good quantum number of the ground state and of every state a sector-preserving probe can reach.

Second, both probes considered below *are* sector preserving. The longitudinal probe $S_\mu^z(\mathbf{q})$ is diagonal in the Ising basis and cannot change a configuration, let alone its transport label; the ring-exchange operator around a *contractible* hexagon returns the configuration to the same sector by construction. The selection rule therefore lives entirely inside the $\mathcal{T} = 0$ block, where $\mathcal{P}$ is exact. (Ring exchange around a non-contractible six-cycle does change the sector; on FCC-32, 96 of the 128 six-cycles wrap the periodic box, and using them is exactly the torus artifact that must be removed first [6].)

Parity of the measurable operators

Table I classifies the operators that appear in real experiments. The classification uses only Eq. (1) and is independent of coupling strength, flux background and cluster.

The content of the table is the statement

$$\langle n | \mathcal{O}_+ | 0 \rangle = 0 \text{ for odd } n, \quad \langle n | \mathcal{O}_- | 0 \rangle = 0 \text{ for even } n, \quad (4)$$

where $\mathcal{O}_\pm$ is any $\mathcal{P}$ -even (odd) operator and $|n\rangle$ is an n -photon state built on a $\mathcal{P}$ -even vacuum. Three corollaries are worth isolating.

(C1) *The one-photon line and the two-photon continuum never appear together.* A neutron measurement of $S^{zz}(\mathbf{q}, \omega)$ in a dipolar QSI contains the single-photon pole and, at higher

TABLE I. Parity of gauge-sector operators under $\mathcal{P}$, and the photon-number change they can produce from a $\mathcal{P}$ -even ground state. $W_{\mathcal{C}} = S_1^+ S_2^- \cdots S_L^-$ is the loop-flip (Wilson) operator, so $W + W^\dagger = 2 \cos B$ and $i(W - W^\dagger) = -2 \sin B$.

operator	$\mathcal{P}$	Δn	experimental channel
$E = S_\mu^z(\mathbf{q})$	—	odd	neutron (dipolar QSI)
$\sin B \sim i(W - W^\dagger)$	—	odd	— (no linear-response probe)
$\cos B \sim W + W^\dagger$	+	even	Raman, THz two-photon
$S_i^+ S_j^- + \text{h.c.}$	+	even	Fleury–Loudon Raman [7]
$S_i^z S_j^z$	+	even	bond striction, ultrasound
$\varepsilon_{ab} \partial H / \partial \varepsilon_{ab}$	+	even	ultrasound, elastic constants
$\hbar \sum_i S_i^z$	—	—	uniform field (breaks $\mathcal{P}$)

energy, the *three*-photon continuum — but not the two-photon continuum, at any order in $J_\pm/J_{zz}$. Conversely a Raman or ultrasound spectrum contains the two-photon continuum with *no* one-photon line to sit on top of it. The frequently invoked worry that a broad two-magnon-like background contaminates the photon line, or that an elastic one-photon peak swamps a weak continuum, is excluded by symmetry.

(C2) $\langle \sin B \rangle = 0$ and $\langle \cos B \rangle \neq 0$ are both consequences of the same rule. The measured hexagon flux $\langle W + W^\dagger \rangle = \pm 1/4$ of the 0- and π -flux backgrounds is $\mathcal{P}$ -even and may be finite; the conjugate quantity is forced to vanish. A nonzero $\langle \sin B \rangle$ in any calculation is a diagnostic of broken $\mathcal{P}$, i.e. of an applied field, a symmetry-breaking approximation, or a finite-size artifact.

(C3) *Photon decay $1 \rightarrow 2$ is exactly forbidden.* The lowest allowed photon-number-changing process is $1 \rightarrow 3$. Together with the vanishing of the polarization bubble below the two-spinon threshold [8], this makes the one-photon line intrinsically sharp on symmetry grounds, not merely kinematically. The leading photon self-interaction is $2 \rightarrow 2$, whose strength is set by the emergent fine structure constant [9].

The zero-momentum theorem

A second, independent superselection rule constrains the long-wavelength limit. On the ice manifold the four sublattice magnetizations $m_\mu = \sum_{i \in \mu} S_i^z$ are *constants* of each electric-flux sector. We have verified on FCC-32 that the map from the 85 transport sectors to the 85 magnetization tuples $(2m_0, 2m_1, 2m_2, 2m_3)$ is a *bijection*: the transport label is not merely determined by the $\mathbf{q} = 0$ magnetizations, it *is* them (the same holds for the 25 sectors of the 16-site cubic cluster). The within-sector spread of every $2m_\mu$ is exactly zero, and the commutator of each $S_\mu^z(\mathbf{q} = 0)$ with an arbitrary sector-block-diagonal operator vanishes identically, against an unmasked control value of 55.9.

Consequently every single-site magnetic operator at $\mathbf{q} = 0$ — the uniform magnetization, the staggered sublattice magnetizations, any ESR or uniform susceptibility vertex — is a *c*-number on the gauge sector and produces *zero* inelastic weight there, to all orders and at any coupling. Only operators that are bilinear in the spins on a bond, and therefore able to move a configuration within a sector, couple to the gauge sector at $\mathbf{q} \rightarrow 0$. This is a symmetry-level explanation of why bond-striction and strain vertices, rather than magnetic dipole vertices, are the correct $\mathbf{q} \rightarrow 0$ probes of the photon [10, 11].

NUMERICAL VERIFICATION

The rule is exact, but it can only be *exhibited* on a cluster whose gauge sector is not contaminated. A periodic pyrochlore cluster is a torus and supports ice-preserving loops that close only through a periodic image; on FCC-32 the shortest is a four-spin process at order $J_\pm^2$, one order *before* the physical hexagon at $|J_\pm|^3$ [6]. Such a process changes the transport sector, so the naive periodic spectrum has no sharp $\mathcal{P}$ classification at $\mathbf{q} = 0$ and no sharp flux sectors. We therefore work with the transport-masked Feshbach map of Ref. [6]: virtual spinon excursions are folded onto the 2970-state FCC-32 ice manifold and all matrix elements between different transport sectors are deleted.

We resolve every level of the masked block by cluster momentum with $P_{\mathbf{q}} = N_t^{-1} \sum_t \chi_{\mathbf{q}}(t)^* T_t$ (projector exact to 4×10^{-13}), and evaluate two channels: the longitudinal weight $|\langle n | S_\mu^z(\mathbf{q}) | 0 \rangle|^2$ and the magnetic weight $|\langle n | \Phi(\mathbf{q}) | 0 \rangle|^2$ with $\Phi(\mathbf{q}) = \sum_h e^{-i\mathbf{q} \cdot \mathbf{r}_h} (W_h + W_h^\dagger)$ summed over the 32 *contractible* hexagons only.

TABLE II. Masked FCC-32, virtual depth $d = 1$. Every level is bright in exactly one channel. E is the longitudinal ($\mathcal{P}$ -odd) weight, $\cos B$ the contractible ring-exchange ($\mathcal{P}$ -even) weight. Energies in units of J_{zz} , measured from the ground state.

$J_{\pm}/J_{zz}$	ω (star)	E weight	$\cos B$ weight
−0.200	0.015678 (L)	1.055	9.9×10^{-26}
	0.025274 (X)	1.1×10^{-27}	1.645
	0.033699 (L)	6.4×10^{-26}	0.201
	0.037928 (X)	0.642	9.2×10^{-26}
+0.046	0.002217 (L)	1.110	1.6×10^{-25}
	0.003612 (X)	1.3×10^{-26}	2.891
	0.004097 (L)	6.2×10^{-27}	0.560
	0.004525 (X)	0.712	9.2×10^{-26}

Table II is the result. The complement of the bright channel lies between 10^{-25} and 10^{-35} throughout, on both sides of the sign of $J_{\pm}$, i.e. in both the π -flux and the 0-flux background. The selection is level by level, not channel by channel: every level the longitudinal probe reaches has all four sublattice channels exactly equal, and every level it misses is at machine zero in all four. An earlier conjecture that different irreducible representations of the little group carry the two channels is thereby excluded — the classification is the global parity, not a spatial label.

Two further checks confirm that the rule is not an accident of the construction. Restricted to the 32 contractible hexagons, the ring exchange conserves the transport sector exactly (out-of-block weight 0.0); including the 96 wrapping six-cycles instead moves 10–28% of the flippable ice weight out of the block, which is the artifact, correctly present. And the longitudinal $\mathbf{q} = 0$ response of Sec. is 1.42×10^{-29} in the masked theory against 0.3126 in the naive periodic theory at $J_{\pm}/J_{zz} = -0.20$ (1.36×10^{-30} against 0.4879 at -0.05).

[PENDING: the $\mathcal{P}$ classification has not yet been checked at $J_{\pm\pm} \neq 0$; the proof in Sec. covers it, but a numerical spot check at the $\text{Ce}_2\text{Hf}_2\text{O}_7$ point $(J_{\pm}, J_{\pm\pm})/J_{zz} = (-0.125, 0.085)$ should be run.]

The missing probe, and a two-photon bound state

Table II contains only one of the two odd operators. Adding the conjugate ring channel

$$\Phi_{-}(\mathbf{q}) = \sum_h e^{-i\mathbf{q}\cdot\mathbf{r}_h} i \left(W_h - W_h^{\dagger} \right) \sim \sin B, \quad (5)$$

which is $\mathcal{P}$ -odd, completes the classification and makes a positive photon assignment possible. A single photon carries both an electric and a magnetic amplitude, so it must be bright in S^z and in Φ_{-} ; a two-photon state must be bright in $\cos B$ and dark in both. The prediction is therefore sharp:

$$\text{levels bright in } S^z \iff \text{bright in } \Phi_{-} \iff \text{odd photon number.} \quad (6)$$

[PENDING: Φ_{-} has not been measured. This is a few-hour run reusing campaign/run_flux_diagnostic.py; see README.]

If Eq. (6) holds, Table II already contains a striking hint. On the fcc reciprocal lattice two L points add to an X point, $(1, 1, 1) + (1, -1, -1) = (2, 0, 0)$, so a non-interacting two-photon state at X built from the lowest L excitation would sit at $2\omega(L)$. Compare:

$J_{\pm}/J_{zz}$	$2\omega(L)$	$\omega_{\text{even}}(X)$	deficit
−0.200	0.031356	0.025274	19.4%
+0.046	0.004434	0.003612	18.5%

The lowest $\mathcal{P}$ -even level at X lies below twice the lowest $\mathcal{P}$ -odd level at L by the same fraction, $\simeq 19\%$, across a $4.3\times$ change of coupling and across the flux-sector boundary. If this is a two-photon bound state, its binding energy is a direct nonperturbative measure of the emergent photon–photon attraction — i.e. of the emergent fine structure constant [9] — and it would appear in Raman and ultrasound as a *sharp line below the onset of the two-photon continuum*.

We flag two reasons this reading is a hypothesis and not yet a result. First, the ratio $\omega_{\text{odd}}(X)/\omega(L)$ is 2.42 and 2.04 at the two couplings, whereas a linear photon dispersion on this cluster would give $|X|/|L| = 2/\sqrt{3} = 1.155$; the FCC-32 spectrum is far from the continuum limit. Second, nothing here is converged in cluster size: between the 16-site cubic cluster and FCC-32 the lowest X excitation moves by a factor 1.5–2.2 and the Γ excitation by 3.5–6.1, and the Γ/X ordering inverts. The selection rule (4) is exact and size independent; the energies it acts on are not. The falsification test is stated in Sec. .

BREAKING THE RULE ON PURPOSE: THE FIELD PROTOCOL

A selection rule that only forbids is hard to use: an absent peak can always be blamed on an intensity. The remedy is that $\mathcal{P}$ has a controlled breaking parameter. A uniform magnetic field enters as $-h \sum_i S_i^z$, which is $\mathcal{P}$ -odd, so at finite h the two families mix. Standard perturbation theory in the field gives, for a channel forbidden at $h = 0$,

$$I_{\text{forbidden}}(h) = \left| \frac{h \langle n | M | n' \rangle \langle n' | \mathcal{O} | 0 \rangle}{E_n - E_{n'}} \right|^2 + O(h^4) \propto h^2, \quad (7)$$

with $M = \sum_i S_i^z$ the uniform magnetization. The protocol is then:

1. At $h = 0$, measure the Raman/ultrasound continuum (even) and the neutron spectrum (odd). The two must not share a feature.
2. Ramp h . A gauge excitation acquires weight in its forbidden channel with an intensity that grows as h^2 from exactly zero, with no threshold.
3. Any conventional excitation — crystal-field level, phonon, magnon of an ordered state, impurity mode — has no reason to have zero weight at $h = 0$ in either channel, and no reason to turn on quadratically.

The signature is the *vanishing at $h = 0$* , which is a property no conventional mode shares. Because M is itself a sector label (Sec.), the field also drives a staircase of transitions between electric-flux sectors; on the 16-site cluster the number of occupied band sectors falls $25 \rightarrow 19 \rightarrow 16 \rightarrow 9$ as h/J_{zz} passes 0.14, 0.20 and 0.28 at $J_{\pm}/J_{zz} = -0.05$, and the naive periodic calculation delays this polarization crossover by a factor 2–3 in h . **[PENDING: the h^2 onset of forbidden weight has not been computed; the field term is not yet wired into the FCC-32 solvers (see README).]**

WHAT EACH EXPERIMENT SEES

Dipolar versus octupolar quantum spin ice

For dipole–octupole doublets [12] the neutron couples only to the dipolar component of the pseudospin. In the exchange-diagonal frame the magnetic dipole is $m \propto \cos \theta \tilde{\tau}^z + \sin \theta \tilde{\tau}^x$. Within the ice manifold, $\tilde{\tau}^x$ flips a spin and therefore leaves the manifold entirely — it creates

TABLE III. Which probe reaches which part of the gauge sector. “—” denotes an exactly vanishing coupling at $h = 0$, not a small one.

probe	vertex	parity	gauge sector, $h = 0$
neutron, dipolar QSI	$E(\mathbf{q})$	—	1-photon, 3-photon
neutron, octupolar QSI	spinon only	—	—
uniform susceptibility, ESR	M	—	— (Sec.)
Raman (Fleury–Loudon)	$S_i \cdot S_j$	+	2-photon continuum
THz two-photon absorption	$\cos B$	+	2-photon
ultrasound / elastic const.	strain–bond	+	2-photon
magnetostriction	strain–bond	+	2-photon

a spinon pair and cannot contribute to the gauge sector. The gauge-sector content of the neutron vertex is thus $\cos \theta \tilde{\tau}^z = \cos \theta E$, a purely $\mathcal{P}$ -odd operator. Two limits follow:

- **Dipolar QSI** ($\cos \theta$ finite, ice built from a dipolar component): neutrons see the one-photon line and the three-photon continuum, and are blind to the two-photon continuum.
- **Octupolar QSI** (ice built from the octupolar component): the neutron vertex has *no* projection on the gauge sector at all. The photon is neutron-dark for a reason stronger than a small form factor, and the even channel is the only access.

The second case is the experimentally pressing one: the Ce pyrochlores $\text{Ce}_2\text{Zr}_2\text{O}_7$ and $\text{Ce}_2\text{Hf}_2\text{O}_7$ are the leading QSI candidates [13–16] and the octupolar assignment is under active discussion. Our statement is then constructive rather than pessimistic: **in octupolar QSI, Raman and ultrasound measure the two-photon continuum, whose onset is at $2\omega_{\min}$ and therefore determines the photon gap that neutrons cannot reach.**

The menu

Table III is the practical summary. Three points deserve emphasis. (i) The even-channel experiments do not compete with the one-photon line; they measure a different object, and they measure it without an elastic background. (ii) At $\mathbf{q} \rightarrow 0$, which is where ultrasound

and Raman work, the magnetic vertices are dead by the theorem of Sec. , so the bond/strain vertex is not merely the largest coupling, it is the only one. (iii) The two-photon onset at $2\omega_{\min}$ is a *lower* edge with a characteristic density-of-states shape, and any sharp line below it is a bound state of emergent light (Sec.).

A warning about small-cluster calibrations

Because the rule is exact, any calculation that violates it is diagnosing its own artifacts. Naive periodic exact diagonalization does violate it: the winding processes mix transport sectors and convert the artifact into observable $\mathbf{q} = 0$ spectral weight (0.31–0.49 of the sum rule on FCC-32, against $\leq 10^{-29}$ after masking). The same artifact inverts the measured flux sign for every $J_{\pm} < 0$, because the winding amplitude $g_4 = 4J_{\pm}^2/J_{zz}$ is even in $J_{\pm}$ while the physical hexagon amplitude $g_6 = 12|J_{\pm}|^3/J_{zz}^2$ is odd. Published identifications calibrated on small periodic clusters should be re-examined against these two diagnostics.

FALSIFICATION TESTS

The paper stands or falls on the following, in decreasing order of cost.

1. **Φ_- complementarity.** Every level bright in S^z must be bright in Φ_- and dark in $\cos B$, and conversely. Failure at machine precision falsifies the identification $\mathcal{P} = C$.
2. **Two-photon bound state.** $2\omega_{\text{odd}}(L) - \omega_{\text{even}}(X)$ must remain positive and approach a finite fraction as the cluster grows, and the even level must be dark in Φ_- . If instead the deficit tracks the finite-size drift of $\omega(X)$ it is not a bound state.
3. **h^2 onset.** Forbidden-channel weight must vanish at $h = 0$ and rise quadratically, with the coefficient set by Eq. (7).
4. **$J_{\pm\pm}$ robustness.** The complementarity must survive the pair-flip term at the Ce-relevant couplings, where S_{tot}^z is no longer conserved.

CONCLUSION

Quantum spin ice has an exact charge conjugation, inherited from the Klein group of the XYZ pseudospin model, under which the emergent electric and magnetic fields both change sign. Photon-number parity is therefore an exact quantum number at zero field, and the spectroscopic probes split into two families that reach disjoint sets of excitations. The one-photon line belongs to neutrons; the two-photon continuum belongs to Raman, THz and ultrasound; a uniform field breaks the rule at order $\hbar^2$ and converts a null test into a positive one. In octupolar quantum spin ice the even channel is the only access to the emergent photon at all. These statements are exact, carry no fitted parameter, and are independent of cluster size — which matters, because the energies they organize are not.

Computations used the transport-masked FCC-32 spectra of Ref. [6].

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